

Robotics 41013: Lab Assignment 2



Total Subject Weight: 35%.

Group Size: 2-3 students (all members MUST contribute meaningfully)

Task	Marked	Week	Date	Worth
Proposal	group	7	21:00 Wednesday 10 September	2/35
Initial Peer Feedback Stage	individual	9	21:00 Wednesday 1 October	3/35
Short Promotion Video	Group	11	21:00 Wednesday 15 October	5/35
Final System Demonstration	Individual	12	Class time Wednesday 22 October	15/35
Final Video	individual	12	21:00 Friday 24 October	10/35
Final Peer Feedback	individual	13	21:00 Wednesday 29 October	<i>Multiplier</i> <i>× Week12_Tasks</i>

Background:

After your success in introducing the robot into their assembly of parts, your company, SafeCo, is looking to expand their product range. SafeCo wants to expand their business and are interested in investigating the possibility of selling small robot systems to customers for use in the home, office or workplace. Management wants your group to spend a few weeks investigating the application of a robot, including integrated safety. SafeCo has made available several¹ real desktop robot manipulators (including their models) for you to test your design on (see below). You need to choose and use one of these. Then model a second new industrial robot arm that is not in the Robotics Toolbox.

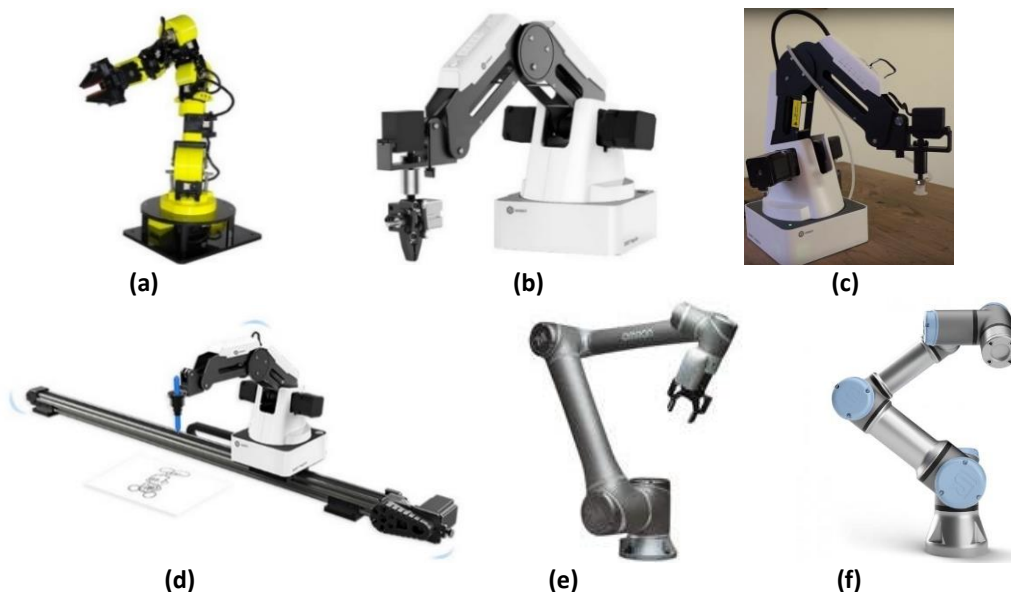


Fig 1. (a) Han's Cute (aka Cyton 300e) with pincer (claw) gripper, (b) Dobot Magician with pincer gripper, (c) Dobot Magician with suction cap gripper, (d) Dobot Linear rail, (e) Omron TM5, (f) Universal Robots UR3e (UTS's grippers not shown)

¹ Drivers for the robots can be found in the Subject Resources page: <https://canvas.uts.edu.au/courses/31794/pages/subject-resources>

SafeCo Pick-and-Place Task:

SafeCo wants you to consider a plausible application for a pick-and-place robot that has a gripper. The application will involve the manipulation of (an) everyday object(s) in a novel way using at least 1 arm that is given and 1 new one that you create. There are likely to be collisions which your system must be able to detect and avoid. So, your group needs to model within your simulated environment (and/or include) at least 3 forms of safety (e.g. barriers and signage, active workspace sensing, collision detection / avoidance and planning), including a functioning e-stop (both in hardware using an Arduino/ESP (or similar and not a PC's keyboard/mouse) and simulated via a GUI which is interacted with via PC's keyboard/mouse), and avoid simulated robot collisions.

Using Real Hardware

All groups need to incorporate a real robot into their demonstration. Using real robots is a rewarding learning experience. We will continue to bring the robots to the weekly class for supervised use. If you have done the lab inductions and safety prerequisites, you will be able to go to the lab unsupervised whenever there is space and time. There will also be special supervised lab sessions available on the Tuesdays of Weeks 11 and 12. Please note that we will only provide technical support for robots in Fig 1. If you happen to have access to a different robot at UTS (e.g., the Sawyer or PR2 in the UTS:RI labs), you are free to use and include that robot in your assignment. But please note that 41013 tutors are unable to officially support these robots.

Progress Tasks - "Journey before Destination"

Throughout the project's journey, there are several important progress tasks. First is the Week 7 Proposal Submission, a group task where you discuss your project ideas with a lead tutor and submit your names, your project idea, robot and code repository details via a Google Form. In Week 9, the Individual Initial Feedback Submission requires each group member to update and improve their A1 SWMS and Risk Assessment. Evidence of regular code repository commits are expected and will be checked, along with a short Spark+ report on group member participation. Finally, in Week 11, a 1-minute group task involves creating a promotional video showcasing the system's purpose, features, and functioning, to be uploaded to YouTube and shared prior to the Week 12 demonstration. These progress tasks collectively monitor and steer the project's development.

Final Submission: Detailed Task Breakdown

Due: Week 12

System Demonstration

Build a simulated model of your chosen arms and environment (based on your chosen scenario).

On the submission day, separate to the videos (submitted the week before), each group needs to actively participate in an online demonstration of their system working both in simulation and, if applicable, on the real robot (students should present video evidence). The simulated system must:

- 1) Include several robot arms working in the same environment:
 - a) **One arm must be an existing robot model from the Robotics Toolbox.**

- b) **Each group member must design and model their own new robot arm** that is *not* already in the Robotics Toolbox.
- c) Each arm must be a 6-DOF (or higher) industrial robot.
It can be a commercially available arm (with documentation) or one you design yourself.
Tip: Choose a robot arm that fits your application and that you can realistically model in simulation.
- d) You should discuss your choice with a tutor to confirm it is suitable for your proposed application.
- e) Each student is marked **only on the robot they personally create and demonstrate**.
Note: If another group member does not complete their own robot, your mark is **not affected**.
- f) A gripper (like the one from Assignment 1) can be included, but it is **not marked**.
- g) The robots do not need to pass objects between them, but they must operate in the same workspace.
- 2) Include a MATLAB/Python graphical user interface (GUI) to interact with the system. The GUI should have advanced “teach” functionality that allows jogging the robot. It should include both individual joint movements (like the Toolbox’s “teach”) plus enable [x,y,z] Cartesian movements. A valid addition is to use a joystick or gamepad to augment and replace mouse-clicking GUI button presses.
- 3) Incorporate a functional e-stop (both simulated and real) that immediately stops operations. Disengaging the e-stop must not immediately resume the robot system but only *permit* resuming (meaning two actions are necessary to resume). For full marks, your system must be able to recover/resume after an e-stop event. Also, a busy “while” loop functionality, to pause everything will be penalised.
- 4) Place the system in a simulated environment that includes safety hardware (e.g., barriers, warning signs/lights/sirens), and if relevant this may be implemented on the real robot as well and augmented with active sensors (BONUS) where data or signals (other than the e-stop) are integrated in the main program.
- 5) Incorporate safety functionality:
 - a) **To react to an asynchronous stop signal by a user.** The system will stop based upon an action from the user (e.g. simulated (or real) sensing of something/someone entering an unsafe zone).
 - b) **To prevent collisions.** When a simulated object (that you make and control) is placed in the path of the robot, it will stop until there is no predicted collision or move to avoid the collision. Active collision avoidance will be marked higher than simply detecting a collision and stopping.
- 6) Ensure the group’s MATLAB/Python code is available on a code repository for tutors to access and download. The code and your understanding will be closely scrutinised during the individual code viva. You are expected to still adhere to a particular code standard (either the one provided, or one chosen by your group).

Final Video

Create a 3-minute (i.e. **exactly** 180-second) professional video. Use real-world footage, simulations, images and diagrams of robot systems to assist in describing your project in detail. Videos should be:

- 1) Interesting to watch (and you should be proud to share with your **professional** engineering network later). It must not be an amalgamation of unrelated and childish meme videos.
- 2) Demonstrate your learning from the subject to solve your novel task,
- 3) Include details about the planning, control, safety, user interfaces, sensing in your system
- 4) Give details of possible extensions, such as sensors, hardware, safety, other robots that would be applicable and to give your system more capabilities

Marking Scheme:

Criteria (task)	Value
Proposal Submission (Due week 7) (group task) Together in a group of two or three, present/discuss your proposal with the lead tutors. Then the group is to submit the following details via a Google Form ² : - Names and students' numbers of group members, - Short project title (less than 10 letters) (will be visible to the class) 100-word description of the intended project. We suggest you use an LLM to ideate and generate this in class. - The robot you intend to use and why - The code repository³ URL for your code (note if your project is private, please invite tutors to join)	2/35
Initial Feedback Submission (Due week 9) (individual task) - SWMS and Risk Assessment assuming the robot is real (even if it is only in simulation) (1st attempt is marked, but resubmits are required until adequate). The same version must be done and submitted as a group. - Evidence of regular commits to repository to show progress - Spark+ report on participation of other group member(s) written from the heart (not by an LLM).	3/35
1-minute promotion video of your specific project (Due week 11) (group task) - Create a 1-minute (exactly 60-second) video promoting your system, include footage and brief details of the system's purpose, its features and the system working. - Upload to YouTube with your student account as a "Unlisted" video. Then, submit the URL in Canvas for marking. Also, share the link on your private discussion board with your allocated tutor and tag all your group members.	5/35
System Demonstration (individual mark based on group task times multiplier from Spark+):	15/35
Each student must demonstration their new robot with completing a specified task within the group's shared environment during an online (in class time) group presentation (10-15 minutes per group): Path plan between several poses and the final joint state given a unique and safe environment (developed by each group) and the simulated model. Use RMRC, visual servoing, collision avoidance and a GUI where appropriate. Creatively use a real robot that mimics and/or enhances the simulation and application. Note: If a student does not create and present their robot they receive 0, and your mark depends on your own robot model, not on other group members' robots.	10
Safety in Demo: - System reacts to user emergency stop action (1) GUI e-stop; (1) Hardware e-stop. (Minus marks if no e-stop). - Trajectory reacts to simulated sensor input (e.g. light curtain) (1) - Trajectory reacts to a forced simulated upcoming collision (1) - Design a simulated environment with a strategically placed models of appropriate safety equipment (1)	5
Final 3-minute video (individual mark based on group task times multiplier from Spark+):	10/35
Demonstrate development and learning to solve a problem with a novel robotic solution	1.5
Footage of your real robot controlled via MATLAB/Python to ROS and integrated with your system	1
Video has professional presentation and is interesting to watch	2

² Google Form to submit group details of Robotics 41013 Lab Assignment #2 <http://goo.gl/c5weiF>

³ We prefer everyone creates and uses a GitHub repository with your UTS email account <https://github.com/>

Includes details of 41013 Robotics learning outcomes such as robot modelling, planning and safety, and user interaction with the system.	4
Robotics sensing: Ideas for applicable sensors that would (1) give the robot more capability; and (2) Improve the system's safety	1
Evidence-based future predictions for robotics in the given scenario	0.5
Final Group Member Peer Feedback Online via Spark+ (individual task) Individual multiplier used to distribute the Demo & Final Video group task marks amongst group members. This task is optional. Only needed if there is a need to distribute group marks in a <u>significantly</u> non-uniform way.	× weight
<u>Total</u>	<u>Out of</u> <u>35</u>
Bonus (1): Up to 2 bonus marks for incorporating additional hardware in your System Demonstration: e.g. for safety, collision detection, an end-effector.	+2/35
Bonus (2): Up to 2 bonus marks for having the system (real or simulated) react to <u>real RGB-D sensor data</u> in your System Demonstration. Require a combination of calibration, mapping, object recognition, image processing.	+2/35

The Final Individual Code Viva is a separate assignment based upon the code written in this assignment. Each student will be examined closely on all the MATLAB/Python code in the repository in a viva (i.e., interview). For full marks, assessors can ask any questions about the code, bugs found or additions to the code. The viva is just to talk about the MATLAB/Python code, so no robots are moved nor are any simulations run. Students may request to be examined on only a portion of the code (which they wrote); however, their marks will be reduced accordingly.

FAQ

1. I'm doing this subject in parallel with Sensor and Control. My group and I are doing a joint project for the two subjects. What are the expectations for 41013? What do I have to demo? Will I get marked on any of the ROS component?

If you are doing this subject in parallel with Sensor and Control and are considering a joint project for the two subjects, then we will expect the following things for this subject:

- A simulated environment that includes your robots and all the other relevant components that you may have (light curtain, barriers, etc), in MATLAB/Python (similar to Assignment 1). For Python students, since Python is part of ROS ecosystem, please note that your primary simulated world must be in Swift (the simulator that comes with the Python robotics toolbox), **NOT** Gazebo.
- You are required to implement planning and controlling of your robots in MATLAB, i.e., you must use inverse kinematic and trajectory generation, as well as RMRC or visual servoing in MATLAB/Python robotics toolbox.
- You can use MoveIt, OpenCV, PCL or other similar ROS packages for other less important components but please note that they won't be marked or questioned in either the demo or your final viva, **except for hardware e-stop and the bonus RGB-D sensor question.**